

CFA Level I

Quantitative Methods

Answer Key

Answer Key with Explanations

220 CFA-style multiple-choice questions
20 questions across each of the 11 learning modules

Grounded in the 2025 CFA Program Curriculum, Level I, Volume 1 - Quantitative Methods. For personal exam preparation and practice.

Quick Answer Grid

Q1: A	Q2: C	Q3: A	Q4: B	Q5: B	Q6: B
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Q19: A	Q20: B	Q21: A	Q22: B	Q23: A	Q24: A
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Q67: A	Q68: A	Q69: A	Q70: A	Q71: A	Q72: A
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Q169: A	Q170: A	Q171: A	Q172: A	Q173: A	Q174: A
Q175: A	Q176: A	Q177: A	Q178: A	Q179: A	Q180: A
Q181: A	Q182: A	Q183: A	Q184: C	Q185: A	Q186: A
Q187: A	Q188: A	Q189: A	Q190: A	Q191: A	Q192: A
Q193: A	Q194: A	Q195: A	Q196: A	Q197: A	Q198: A
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Q205: A	Q206: A	Q207: A	Q208: A	Q209: A	Q210: A
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Module 1: Rates and Returns

20 questions

Q1. Answer: A - Required rate of return, discount rate, and opportunity cost

Rationale: The curriculum states an interest rate r can be interpreted as (1) a required rate of return, (2) a discount rate, or (3) an opportunity cost.

Q2. Answer: C - Equity risk premium

Rationale: The interest rate equals the real risk-free rate plus inflation, default, liquidity, and maturity premiums. The 'equity risk premium' is not part of this decomposition.

Q3. Answer: A - real risk-free rate and the inflation premium

Rationale: The nominal risk-free rate is approximated as the real risk-free rate plus the inflation premium.

Q4. Answer: B - 12.0%

Rationale: $HPR = (54 - 50 + 2)/50 = 6/50 = 12.0\%$. Total return = income yield + capital gain.

Q5. Answer: B - Time-weighted return

Rationale: The time-weighted return is not sensitive to the timing/amount of cash flows and is preferred for evaluating portfolio managers. The money-weighted return (an IRR) is sensitive to cash flow timing.

Q6. Answer: B - external cash flows are generally outside the manager's control

Rationale: Cash withdrawals/additions are generally outside the manager's control, so the time-weighted return - which removes their effect - is preferred for performance evaluation.

Q7. Answer: B - internal rate of return on a portfolio accounting for all cash flows

Rationale: The money-weighted return is the IRR that reflects the actual return earned after accounting for the value and timing of all cash flows.

Q8. Answer: B - -13.4%

Rationale: Geometric mean = $(1.50 \times 0.50)^{(1/2)} - 1 = (0.75)^{0.5} - 1 = -13.4\%$. The arithmetic mean would be 0%.

Q9. Answer: B - always less

Rationale: For any non-constant series, the geometric mean is always less than the arithmetic mean; they are equal only if all returns are identical.

Q10. Answer: A - Harmonic mean

Rationale: The harmonic mean is appropriate for averaging ratios/rates (e.g., cost averaging, P/E) and reduces the impact of large outliers.

Q11. Answer: B - excluding a stated small percentage of the lowest and highest values

Rationale: A trimmed mean excludes a small stated percentage of the lowest and highest values. A winsorized mean replaces (not excludes) the extremes.

Q12. Answer: B - replaces the extreme observations with specified boundary values

Rationale: A winsorized mean replaces extreme values with specified boundary (percentile) values rather than excluding them.

Q13. Answer: B - 12.55%

Rationale: Annualized = $(1.03)^4 - 1 = 1.1255 - 1 = 12.55\%$.

Q14. Answer: A - compounded continuously

Rationale: With the number of compounding periods approaching infinity, interest is compounded continuously, using e^r .

Q15. Answer: A - 5.83%

Rationale: Continuously compounded return = $\ln(1 + HPR) = \ln(1.06) = 5.83\%$.

Q16. Answer: B - prior to deduction of managerial and administrative expenses not directly related to return generation

Rationale: Gross return is measured before deducting managerial/administrative expenses (those not directly related to generating returns) and is appropriate for comparing manager performance.

Q17. Answer: A - gross return less managerial and administrative expenses

Rationale: Net return = gross return less managerial and administrative expenses; it better reflects what the investor actually earned.

Q18. Answer: B - any allowance for taxes on dividends, interest, and realized gains

Rationale: After-tax nominal return = total return minus allowances for taxes on dividends, interest, and realized capital gains.

Q19. Answer: A - comparing returns across time periods when inflation rates vary

Rationale: Real returns are useful for comparing returns across time periods (because inflation varies) and across asset classes with different taxation.

Q20. Answer: B - amplify both the portfolio's gains and losses

Rationale: Leverage amplifies a portfolio's gains and losses.

Module 2: Time Value of Money in Finance

20 questions

Q21. Answer: A - promised coupon payments and its par value

Rationale: A bond's price equals the present value of its promised coupons plus the present value of its par value.

Q22. Answer: B - the par value only

Rationale: Discount (zero-coupon) bonds have no coupons, so the price reflects only the present value of the par value.

Q23. Answer: A - the stock's expected future dividends in perpetuity

Rationale: A stock's value should reflect the present value of its expected future dividends in perpetuity.

Q24. Answer: A - expected growth pattern of future dividends

Rationale: Dividend-based valuation models are classified by the assumed dividend growth pattern: no growth, constant growth, or changing growth.

Q25. Answer: B - 33.33

Rationale: $V = D1/(r - g) = 2.00/(0.10 - 0.04) = 2.00/0.06 = 33.33$.

Q26. Answer: A - perpetuity: D / r

Rationale: With no growth, the stock is a level perpetuity valued at D/r .

Q27. Answer: A - $(D1 / P0) + g$

Rationale: Rearranging $P0 = D1/(r - g)$ gives $r = D1/P0 + g$; the required return equals the dividend yield plus the growth rate.

Q28. Answer: A - 5.0%

Rationale: $g = r - D1/P0 = 0.09 - 1.60/40 = 0.09 - 0.04 = 5.0\%$.

Q29. Answer: A - rate that equates the present value of cash flows to the price

Rationale: The implied return (yield-to-maturity) is the discount rate that sets the present value of the bond's promised cash flows equal to its price.

Q30. Answer: A - occur at the same point in time

Rationale: Cash flow additivity applies when cash flows occur at the same point in time, allowing streams to be added/compared.

Q31. Answer: A - the no-arbitrage condition holds

Rationale: Cash flow additivity underpins no-arbitrage pricing: economically equivalent assets must have the same price.

Q32. Answer: A - economically equivalent assets, even with differing cash flow streams

Rationale: Cash flow additivity confirms economically equivalent assets have the same price even if their cash flow streams differ.

Q33. Answer: A - the cash flow additivity / no-arbitrage principle

Rationale: Implied forward rates are derived from spot rates using no-arbitrage (cash flow additivity) relationships.

Q34. Answer: B - $FV = PV \times e^{(rt)}$

Rationale: Under continuous compounding, $FV = PV \times e^{(rt)}$.

Q35. Answer: B - par value (1,000)

Rationale: When the coupon rate equals the market discount rate, the bond prices at par.

Q36. Answer: B - at a discount

Rationale: When the required return exceeds the coupon rate, the present value of cash flows falls below par - the bond trades at a discount.

Q37. Answer: A - a higher stock value

Rationale: A higher g reduces the denominator $(r - g)$, increasing the stock's value, all else equal.

Q38. Answer: C - the skewness of a return distribution

Rationale: Cash flow additivity is used for forward rates, forward FX rates, and option values. Skewness is a statistical moment unrelated to no-arbitrage pricing.

Q39. Answer: A - dividends are expected to grow at a high rate then settle to a stable rate

Rationale: A changing-growth (multi-stage) model is used when dividends grow at one rate initially and a different (often lower, stable) rate later.

Q40. Answer: A - $PV = FV \times e^{(-rt)}$

Rationale: Under continuous discounting, $PV = FV \times e^{(-rt)}$.

Module 3: Statistical Measures of Asset Returns

20 questions

Q41. Answer: B - Median

Rationale: The median is the middle value and is not influenced by extreme values, making it useful for skewed distributions.

Q42. Answer: C - mode

Rationale: The mode (most frequently observed value) is the only measure of central tendency usable with nominal/categorical data.

Q43. Answer: B - bimodal

Rationale: A distribution with two modes is bimodal.

Q44. Answer: B - quantiles

Rationale: These location parameters that divide a distribution into equal parts are collectively called quantiles.

Q45. Answer: A - interquartile range (Q3 - Q1)

Rationale: The box represents the interquartile range - the difference between the third and first quartiles.

Q46. Answer: A - uses information from only two observations

Rationale: The range uses only the maximum and minimum values - just two observations - limiting its usefulness.

Q47. Answer: B - $n - 1$

Rationale: Sample variance uses a divisor of $n - 1$ (degrees of freedom) to provide an unbiased estimate.

Q48. Answer: B - absolute deviations of observations from the mean

Rationale: The MAD is the average of the absolute deviations of observations from the mean.

Q49. Answer: B - target downside deviation

Rationale: Target downside (semi)deviation measures the risk of outcomes below a given target return.

Q50. Answer: A - the standard deviation divided by the mean

Rationale: $CV = \text{standard deviation} / \text{mean}$; it is a scale-free measure of relative dispersion.

Q51. Answer: B - is a scale-free measure allowing direct comparison across datasets

Rationale: By dividing dispersion by the mean, CV becomes scale-free (no units), enabling direct comparisons across datasets.

Q52. Answer: A - a lower CV (less risk per unit of return)

Rationale: $CV = SD/\text{mean}$. With equal SD and a higher mean, Portfolio A has a lower CV, i.e., less risk per unit of return.

Q53. Answer: A - positive skewness

Rationale: Positive (right) skewness: frequent small losses and a few extreme gains; the long tail is on the right.

Q54. Answer: B - negatively skewed

Rationale: Negative (left) skewness: frequent small gains and a few extreme losses; the long tail is on the left.

Q55. Answer: C - 3

Rationale: The kurtosis of a normal distribution is 3; excess kurtosis ($\text{kurtosis} - 3$) is therefore 0.

Q56. Answer: A - leptokurtic (fat-tailed)

Rationale: A fat-tailed distribution is leptokurtic; a thin-tailed distribution is platykurtic.

Q57. Answer: B - fat-tailed relative to normal

Rationale: Positive excess kurtosis ($\text{kurtosis} > 3$) indicates fatter tails than the normal distribution (leptokurtic).

Q58. Answer: B - more likely than under a normal distribution

Rationale: Fat tails (leptokurtosis) mean extreme outcomes occur more frequently than a normal distribution would predict - an important risk consideration.

Q59. Answer: B - -1 and +1

Rationale: Correlation, a standardized measure of linear association, ranges from -1 to +1.

Q60. Answer: B - little or no linear relationship

Rationale: A correlation near zero indicates little or no linear relationship between the two variables.

Module 4: Probability Trees and Conditional Expectations

20 questions

Q61. Answer: A - probability-weighted average of the possible outcomes

Rationale: $E(X)$ is the probability-weighted average of the random variable's possible outcomes.

Q62. Answer: B - $E\{[X - E(X)]^2\}$

Rationale: Variance is the probability-weighted average of squared deviations from the mean: $\sigma^2(X) = E\{[X - E(X)]^2\}$.

Q63. Answer: A - 4.0%

Rationale: $E(R) = 0.60(10\%) + 0.40(-5\%) = 6\% - 2\% = 4.0\%$.

Q64. Answer: B - measured in the same units as the variable

Rationale: Standard deviation is the positive square root of variance and is expressed in the same units as the variable, aiding interpretation.

Q65. Answer: A - the results of two or more events and their probabilities

Rationale: A probability tree visually illustrates the outcomes of two or more events along with conditional and joint probabilities.

Q66. Answer: A - $P(AB) / P(B)$, with $P(B)$ not equal to 0

Rationale: By definition, $P(A|B) = P(AB)/P(B)$, provided $P(B)$ is not zero.

Q67. Answer: A - $P(A|B) = P(A)$

Rationale: Independence means the occurrence of B does not change the probability of A: $P(A|B) = P(A)$, equivalently $P(AB) = P(A)P(B)$.

Q68. Answer: A - the sum of $P(A | S_i) \times P(S_i)$ across all scenarios

Rationale: The total probability rule: $P(A) = \sum_i P(A|S_i)P(S_i)$ for mutually exclusive and exhaustive scenarios.

Q69. Answer: A - update prior probabilities based on new information

Rationale: Bayes' formula provides a rational method to update (revise) prior probabilities as new information arrives.

Q70. Answer: A - probability of the new information given the event to the unconditional probability of the new information

Rationale: Updated probability = $[P(\text{information}|\text{event})/P(\text{information})] \times \text{prior probability of the event}$.

Q71. Answer: A - using conditional probabilities given scenario S

Rationale: $E(X|S) = \sum P(X_i|S) \times X_i$ - a probability-weighted average using conditional probabilities given S.

Q72. Answer: A - cannot occur together and cover all possibilities

Rationale: Mutually exclusive = cannot occur simultaneously; exhaustive = together cover all possible outcomes.

Q73. Answer: A - $a \times E(X)$

Rationale: Expectation is a linear operator: $E(aX) = a \times E(X)$.

Q74. Answer: A - +6.0%

Rationale: $E(R) = 0.30(-8\%) + 0.70(12\%) = -2.4\% + 8.4\% = 6.0\%$.

Q75. Answer: A - revising the probability of a state given a new signal or observation

Rationale: Bayes' formula updates the probability of an underlying state given a newly observed signal (e.g., an earnings surprise).

Q76. Answer: A - 0

Rationale: Mutually exclusive events cannot occur together, so their joint probability $P(A \text{ and } B) = 0$.

Q77. Answer: A - probability-weighting the squared deviations of each terminal outcome from $E(X)$

Rationale: Variance = probability-weighted sum of squared deviations of each outcome from the expected value $E(X)$.

Q78. Answer: A - 0.40

Rationale: $P(A|B) = P(AB)/P(B) = 0.2/0.5 = 0.40$.

Q79. Answer: A - total probability rule

Rationale: The unconditional probability of the information is found via the total probability rule across all states.

Q80. Answer: A - non-negative

Rationale: Standard deviation is the positive square root of variance and is therefore always non-negative.

Module 5: Portfolio Mathematics

20 questions

Q81. Answer: A - a weighted average of the expected returns on the component securities

Rationale: $E(R_p)$ is the weighted average of the component securities' expected returns, using portfolio weights.

Q82. Answer: A - 9.6%

Rationale: $E(R_p) = 0.60(8\%) + 0.40(12\%) = 4.8\% + 4.8\% = 9.6\%$.

Q83. Answer: A - covariances

Rationale: Portfolio variance reflects individual asset variances and the covariances (co-movement) between asset returns.

Q84. Answer: B - negative, zero, or positive

Rationale: Covariance can be negative (inverse relationship), zero (unrelated), or positive (direct relationship). Correlation, by contrast, is bounded by -1 and +1.

Q85. Answer: A - $2 w_1 w_2 \text{Cov}(R_1, R_2)$

Rationale: Two-asset portfolio variance = $w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \text{Cov}(R_1, R_2)$.

Q86. Answer: A - the product of the two standard deviations

Rationale: Correlation = $\text{Cov}(R_i, R_j) / (\sigma_i \times \sigma_j)$.

Q87. Answer: C - -1

Rationale: The lower the correlation, the greater the diversification benefit; a correlation of -1 offers the maximum risk reduction.

Q88. Answer: A - the weighted average of the two standard deviations

Rationale: With perfect positive correlation, no diversification benefit exists, and the portfolio SD equals the weighted average of the component SDs.

Q89. Answer: A - $P(X, Y) = P(X)P(Y)$

Rationale: Independence requires $P(X, Y) = P(X)P(Y)$ for all values.

Q90. Answer: A - the product of their expected values

Rationale: For independent variables, $E(XY) = E(X)E(Y)$.

Q91. Answer: A - probability that a portfolio's value will fall below a minimum acceptable level

Rationale: Shortfall risk is the probability that the portfolio return/value falls below a specified minimum (threshold) level.

Q92. Answer: A - maximizes the safety-first ratio (SFRatio)

Rationale: Roy's safety-first criterion chooses the portfolio with the highest safety-first ratio, minimizing the probability of returns below the threshold.

Q93. Answer: A - $[E(R_p) - RL] / \sigma_p$

Rationale: $\text{SFRatio} = [E(R_p) - RL] / \sigma_p$, where RL is the threshold (minimum acceptable) return.

Q94. Answer: A - 0.50

Rationale: $\text{SFRatio} = (10\% - 2\%) / 16\% = 8 / 16 = 0.50$.

Q95. Answer: A - probability of returns below the threshold level

Rationale: A higher SFRatio corresponds to a lower probability of returns falling below the threshold (under normality).

Q96. Answer: A - variance

Rationale: $\text{Cov}(R, R) = \text{Var}(R)$; an asset's covariance with itself is its own variance.

Q97. Answer: A - reduce the portfolio's overall risk

Rationale: Low or negative correlation enhances diversification, reducing overall portfolio risk.

Q98. Answer: A - $P(R_{A,i}, R_{B,j}) \times (R_{A,i} - E[R_A]) \times (R_{B,j} - E[R_B])$

Rationale: $\text{Cov}(R_A, R_B) = \sum \text{over } i, j \text{ of } P(R_{A,i}, R_{B,j}) (R_{A,i} - E[R_A]) (R_{B,j} - E[R_B])$ - probability-weighted cross products of deviations.

Q99. Answer: A - 0.55

Rationale: Roy's safety-first criterion selects the portfolio with the highest SFRatio - here, 0.55.

Q100. Answer: A - The expected returns of the assets and their portfolio weights

Rationale: Portfolio expected return is the weighted average of component expected returns, requiring the assets' expected returns and weights.

Module 6: Simulation Methods

20 questions

Q101. Answer: A - the natural logarithm of Y is normally distributed

Rationale: Y is lognormal if $\ln(Y)$ is normally distributed ('the log is normal').

Q102. Answer: A - bounded below by zero and skewed to the right

Rationale: Asset prices cannot be negative; the lognormal distribution is bounded below by 0 and is right-skewed, matching price behavior.

Q103. Answer: A - skewed to the right (long right tail)

Rationale: The lognormal distribution is skewed to the right with a long right tail.

Q104. Answer: A - The normal distribution

Rationale: The normal distribution is often a good approximation for returns, while the lognormal is used for prices.

Q105. Answer: A - Lognormal

Rationale: If continuously compounded returns are normally distributed, prices are lognormally distributed.

Q106. Answer: A - value securities with complex features where no analytic pricing formula exists

Rationale: Monte Carlo simulation is used to estimate risk/return and value complex securities (e.g., with embedded options) lacking closed-form solutions.

Q107. Answer: A - generating a large number of random samples from specified probability distributions

Rationale: Monte Carlo generates many random samples from specified distribution(s) to obtain the likelihood of a range of outcomes.

Q108. Answer: A - its results are only as good as the assumptions and distributions specified

Rationale: Monte Carlo output depends entirely on the input assumptions/distributions chosen ('garbage in, garbage out').

Q109. Answer: A - repeatedly drawing samples from the observed sample, treating it as the population

Rationale: Bootstrapping treats the observed sample as if it were the population and repeatedly resamples from it.

Q110. Answer: A - a random sample offers a good representation of the population

Rationale: Because a random sample represents the population well, resampling from it mimics sampling from the population.

Q111. Answer: A - resamples from observed historical data rather than a specified theoretical distribution

Rationale: Bootstrapping draws from observed data; Monte Carlo draws from specified theoretical distributions.

Q112. Answer: A - Analytical (closed-form) methods

Rationale: Monte Carlo complements analytical methods, providing estimates where closed-form (analytical) solutions are unavailable.

Q113. Answer: A - expected payoff and hence its value

Rationale: Simulating many price paths yields a distribution of payoffs whose average (discounted) approximates the option's value.

Q114. Answer: C - guaranteeing future investment returns

Rationale: Bootstrapping estimates sampling distributions (e.g., standard errors) and supports historical simulation, but cannot guarantee future returns.

Q115. Answer: A - the initial price times e raised to the return

Rationale: $\text{Price} = P_0 \times e^{(\text{return})}$; exponentiating a normal return produces a lognormal price.

Q116. Answer: A - assessing the risk and distribution of potential results

Rationale: Monte Carlo characterizes the distribution and risk of outcomes rather than producing a single certain answer.

Q117. Answer: A - Black-Scholes-Merton

Rationale: The Black-Scholes-Merton model assumes the underlying asset price is lognormally distributed.

Q118. Answer: A - negative values

Rationale: The lognormal distribution is bounded below by zero, so it cannot take negative values - suitable for asset prices.

Q119. Answer: A - Specifying the probability distribution(s) for the risk factors

Rationale: A Monte Carlo simulation requires specifying the probability distribution(s) of the underlying risk factors before generating random samples.

Q120. Answer: A - the same size as the original observed sample

Rationale: Bootstrap repeatedly draws samples of the same size as the original sample (with replacement).

Module 7: Estimation and Inference

20 questions

Q121. Answer: A - Simple random sampling

Rationale: Probability sampling includes simple random and stratified random sampling. Convenience and judgmental sampling are non-probability methods.

Q122. Answer: A - dividing the population into subgroups and randomly sampling from each

Rationale: Stratified random sampling divides the population into strata and draws random samples from each subgroup.

Q123. Answer: A - a sample statistic and the population parameter it estimates

Rationale: Sampling error is the difference between the observed value of a statistic and the parameter it is intended to estimate.

Q124. Answer: A - allow quick and low-cost data collection

Rationale: Non-probability methods are quick and low-cost and can apply expert judgment, but carry a significant risk of being non-representative.

Q125. Answer: A - non-representative of the population

Rationale: Non-probability sampling does not give every population member an equal chance of selection, risking a non-representative sample.

Q126. Answer: A - normal with mean μ and variance σ^2/n

Rationale: The CLT: for large n , the sample mean is approximately normal with mean μ and variance σ^2/n , regardless of the population distribution.

Q127. Answer: A - regardless of the shape of the underlying population distribution

Rationale: The CLT applies to any population distribution with finite variance, provided the sample size is large.

Q128. Answer: A - the square root of n

Rationale: Standard error of the mean = $\sigma / \sqrt{n}$ (estimated using $s/\sqrt{n}$ when σ is unknown).

Q129. Answer: A - 2.0

Rationale: $SE = \sigma/\sqrt{n} = 20/\sqrt{100} = 20/10 = 2.0$.

Q130. Answer: A - decrease

Rationale: Because $SE = \sigma/\sqrt{n}$, a larger n reduces the standard error, improving the precision of the estimate.

Q131. Answer: A - population mean

Rationale: The CLT establishes that the mean of the sampling distribution of the sample mean equals the population mean μ .

Q132. Answer: A - estimate the sampling distribution of a statistic

Rationale: Bootstrap and jackknife are resampling methods used to estimate the sampling distribution (e.g., standard error) of a statistic.

Q133. Answer: A - uses computer simulation rather than an analytical formula such as a z- or t-statistic

Rationale: The bootstrap uses computer-based resampling instead of an analytical formula, and works for complicated estimators.

Q134. Answer: A - accuracy and wide applicability to complicated estimators

Rationale: The bootstrap can offer accuracy advantages and applies widely (e.g., asset allocation simulations, benchmark comparisons).

Q135. Answer: A - Convenience sampling

Rationale: Convenience sampling selects observations that are easiest to obtain - a non-probability method.

Q136. Answer: A - an equal chance of being selected

Rationale: Simple random sampling gives every population member an equal probability of selection.

Q137. Answer: A - precision (variability) of the sample mean as an estimate of the population mean

Rationale: The standard error quantifies how much the sample mean is expected to vary from the population mean - i.e., the estimate's precision.

Q138. Answer: A - repeatedly drawing samples of the same size from the original sample

Rationale: Bootstrapping repeatedly draws samples of the same size as the original (with replacement) to build a sampling distribution.

Q139. Answer: A - the researcher's expertise to select the sample

Rationale: Judgmental sampling uses the researcher's expert judgment to select sample members - a non-probability method.

Q140. Answer: A - reduce sampling error and increase estimate precision

Rationale: Larger samples generally reduce sampling error and produce more precise estimates of population parameters.

Module 8: Hypothesis Testing

20 questions

Q141. Answer: A - State the hypotheses

Rationale: The steps begin with (1) stating the hypotheses, then identifying the test statistic, specifying significance, stating the decision rule, computing, and deciding.

Q142. Answer: A - rejects a true null hypothesis

Rationale: A Type I error (false positive) is rejecting a null hypothesis that is actually true.

Q143. Answer: A - fails to reject a false null hypothesis

Rationale: A Type II error (false negative) is failing to reject a null hypothesis that is actually false.

Q144. Answer: A - a Type I error

Rationale: The significance level (α) is the probability of committing a Type I error.

Q145. Answer: A - the probability of correctly rejecting a false null hypothesis

Rationale: Power = $1 - P(\text{Type II error})$; it is the probability of correctly rejecting a false null hypothesis.

Q146. Answer: A - level of significance

Rationale: Confidence level = $1 - \text{significance level } (\alpha)$.

Q147. Answer: A - t-statistic

Rationale: When the population variance is unknown, the t-statistic is theoretically correct for testing the population mean.

Q148. Answer: A - the basis for deciding whether to reject the null hypothesis

Rationale: The test statistic is computed from the sample and compared to a critical value to decide whether to reject the null.

Q149. Answer: A - t-test based on pooling the two samples' observations

Rationale: For independent samples from normal populations with unknown but equal variances, a pooled-variance t-test is used.

Q150. Answer: A - paired comparisons (mean differences) test

Rationale: Dependent (related) samples call for a paired comparisons test (a test of mean differences).

Q151. Answer: A - chi-square with $n - 1$ degrees of freedom

Rationale: A test of a single population variance uses a chi-square statistic with $n - 1$ degrees of freedom.

Q152. Answer: A - an F-statistic (ratio of sample variances)

Rationale: Comparing two population variances uses an F-test - the ratio of the two sample variances.

Q153. Answer: A - $n_1 - 1$ and $n_2 - 1$

Rationale: The F-test uses two degrees-of-freedom parameters: $n_1 - 1$ (numerator) and $n_2 - 1$ (denominator).

Q154. Answer: A - concerns a population parameter or relies on specific distributional assumptions

Rationale: Parametric tests concern a parameter and/or rely on specific distributional assumptions.

Q155. Answer: A - reject the null hypothesis

Rationale: When the test statistic falls in the rejection region (beyond the critical value), the null hypothesis is rejected.

Q156. Answer: A - decrease the probability of a Type I error but increase the probability of a Type II error

Rationale: Lowering α reduces Type I error probability but, all else equal, raises the probability of a Type II error.

Q157. Answer: A - smallest significance level at which the null hypothesis can be rejected

Rationale: The p-value is the smallest level of significance at which the null can be rejected; reject if $p\text{-value} < \alpha$.

Q158. Answer: A - is testing for a difference in a specific direction

Rationale: A one-sided test is used when the hypothesis specifies a direction (greater than or less than).

Q159. Answer: A - statement to be tested, often 'no effect' or 'no difference'

Rationale: The null hypothesis is the statement to be tested (often 'no effect'); the alternative is what is concluded if the null is rejected.

Q160. Answer: A - accepting a higher probability of a Type I error

Rationale: For a fixed sample size, there is a trade-off: increasing power (reducing Type II error) generally requires accepting a higher Type I error probability (higher alpha).

Module 9: Parametric and Non-Parametric Tests of Independence

20 questions

Q161. Answer: A - $H_0: \theta = \theta_0$ versus $H_a: \theta \neq \theta_0$

Rationale: A two-sided test: $H_0: \theta = \theta_0$ versus $H_a: \theta \neq \theta_0$.

Q162. Answer: A - data do not meet distributional assumptions or are given in ranks

Rationale: Nonparametric tests are used when distributional assumptions are not met, outliers exist, data are ranks, or the hypothesis does not concern a parameter.

Q163. Answer: A - $n - 2$

Rationale: The t-test for a correlation coefficient has $n - 2$ degrees of freedom.

Q164. Answer: A - $H_0: \rho = 0$

Rationale: The common test asks whether any relationship exists, testing $H_0: \rho = 0$ (no correlation).

Q165. Answer: A - ranks of the two variables within their samples

Rationale: The Spearman rank correlation is computed on the ranks of the two variables, not their raw values.

Q166. Answer: A - data do not meet the assumptions required for the parametric correlation test

Rationale: Spearman rank correlation (nonparametric) is useful when parametric assumptions (e.g., normality) are not met or with ranked/outlier-prone data.

Q167. Answer: A - Chi-square

Rationale: A chi-square test statistic is used to test the independence of two categorical variables using a contingency table.

Q168. Answer: A - the assumption of independence

Rationale: The chi-square test of independence compares observed frequencies to those expected if the two variables were independent.

Q169. Answer: A - displays the joint frequencies of two categorical variables

Rationale: A contingency table presents the joint frequencies (counts) of observations across categories of two variables.

Q170. Answer: A - (row total x column total) / grand total

Rationale: Expected cell frequency = (row total x column total) / grand total under the independence assumption.

Q171. Answer: A - makes minimal assumptions about the population distribution

Rationale: Nonparametric tests make minimal (or no) assumptions about the underlying population distribution.

Q172. Answer: A - concerns a parameter or relies on specific distributional assumptions

Rationale: Parametric tests concern a population parameter and/or assume a specific distribution; nonparametric tests do not.

Q173. Answer: A - not independent (i.e., related)

Rationale: Rejecting the null ($\chi^2 > \text{critical}$) indicates the variables are not independent - there is a relationship.

Q174. Answer: A - The presence of significant outliers in small samples

Rationale: Nonparametric tests are preferred with outliers, ranked data, small samples, or when distributional assumptions are violated.

Q175. Answer: A - $H_a: \rho > 0$

Rationale: A right-side (one-sided) test: $H_0: \rho \leq 0$ versus $H_a: \rho > 0$.

Q176. Answer: A - increases

Rationale: A larger n increases the t-statistic for correlation (the formula multiplies r by $\sqrt{(n-2)/(1-r^2)}$), making significance easier to detect.

Q177. Answer: A - $(\text{observed} - \text{expected})^2 / \text{expected}$

Rationale: The chi-square statistic = sum over cells of $(\text{observed} - \text{expected})^2 / \text{expected}$.

Q178. Answer: A - chi-square test of independence

Rationale: Relationships between two categorical variables are tested with a chi-square test of independence using a contingency

table.

Q179. Answer: A - the observed correlation could plausibly be due to chance

Rationale: Failing to reject $H_0: \rho = 0$ means the evidence is insufficient to conclude a real (non-chance) linear relationship exists.

Q180. Answer: A - is the result of chance

Rationale: A significance test of correlation assesses whether the observed relationship between two variables is merely the result of chance.

Module 10: Simple Linear Regression

20 questions

Q181. Answer: A - dependent variable

Rationale: The dependent variable is the one being explained; the independent variable is used to explain it.

Q182. Answer: A - Ordinary least squares

Rationale: Simple linear regression uses ordinary least squares (OLS) to estimate the intercept and slope parameters.

Q183. Answer: A - squared vertical distances between observations and the regression line

Rationale: OLS minimizes the sum of squared residuals (squared vertical distances between actual Y values and the fitted line).

Q184. Answer: C - Multicollinearity

Rationale: The four assumptions are linearity, homoskedasticity, independence, and normality. Multicollinearity relates to multiple regression, not simple regression.

Q185. Answer: A - variance of the residuals is constant across observations

Rationale: Homoskedasticity means the residual variance is constant across all observations.

Q186. Answer: A - X-Y pairs are uncorrelated across observations

Rationale: Independence requires that observations (X-Y pairs) be uncorrelated; patterns like seasonality suggest autocorrelation, violating it.

Q187. Answer: A - residuals be normally distributed

Rationale: Normality requires the residuals (not the raw data) to be normally distributed.

Q188. Answer: A - explained variation (SSR) plus unexplained variation (SSE)

Rationale: $SST = SSR$ (sum of squares regression, explained) + SSE (sum of squares error, unexplained).

Q189. Answer: A - proportion of the total variation in Y explained by the regression

Rationale: $R\text{-squared} = SSR/SST$ is the fraction of total variation in the dependent variable explained by the independent variable.

Q190. Answer: A - square of the correlation coefficient between X and Y

Rationale: In simple linear regression, $R\text{-squared}$ equals the square of the correlation between X and Y.

Q191. Answer: A - standard deviation of the regression residuals

Rationale: The SEE measures the dispersion of actual Y values around the fitted regression line (the standard deviation of residuals).

Q192. Answer: A - fits the data more closely

Rationale: A smaller SEE means observations lie closer to the regression line - a better fit.

Q193. Answer: A - evaluate the overall fit and significance of the regression

Rationale: ANOVA decomposes variation (SST into SSR and SSE) and supports the F-test of overall regression significance.

Q194. Answer: A - regression as a whole is statistically significant (slope differs from zero)

Rationale: The F-test assesses overall fit; in simple regression it tests whether the slope coefficient differs from zero.

Q195. Answer: A - t-test on the estimated slope

Rationale: A t-test (estimated coefficient divided by its standard error) is used to test whether the slope differs from zero.

Q196. Answer: A - change in Y for a one-unit change in X

Rationale: The slope b_1 is the expected change in the dependent variable Y for a one-unit change in the independent variable X.

Q197. Answer: A - expected value of Y when X equals zero

Rationale: The intercept b_0 is the expected value of the dependent variable when the independent variable equals zero.

Q198. Answer: A - uncertainty in the regression model and the residual variability

Rationale: A prediction interval incorporates the uncertainty of the estimated regression and the variability of the residuals around the prediction.

Q199. Answer: A - detect violations of the regression assumptions

Rationale: Residual plots help reveal violations such as heteroskedasticity, non-linearity, or autocorrelation.

Q200. Answer: A - independence (suggesting autocorrelation)

Rationale: A systematic pattern such as seasonality in residuals suggests autocorrelation, violating the independence assumption.

Module 11: Introduction to Big Data Techniques

20 questions

Q201. Answer: A - volume, velocity, and variety

Rationale: Big Data is characterized by the three Vs: volume, velocity, and variety.

Q202. Answer: A - veracity

Rationale: A fourth V - veracity (data quality/reliability) - becomes important when Big Data is used for inference or prediction.

Q203. Answer: A - technology-driven innovation in the financial services industry

Rationale: Fintech refers to technology-driven innovation in the design and delivery of financial services and products.

Q204. Answer: A - Individuals, business processes, and sensors

Rationale: Main sources of alternative data are data generated by individuals, business processes, and sensors.

Q205. Answer: A - performing tasks that traditionally required human intelligence

Rationale: AI systems perform tasks traditionally requiring human intelligence at levels comparable or superior to humans.

Q206. Answer: A - 'find the pattern, apply the pattern'

Rationale: ML seeks to extract knowledge by learning from examples - 'find the pattern, apply the pattern'.

Q207. Answer: A - Supervised learning

Rationale: Main types of ML include supervised learning, unsupervised learning, and deep learning.

Q208. Answer: A - labeled data (inputs paired with known outputs)

Rationale: Supervised learning uses labeled data - inputs paired with known target outputs - to learn to predict.

Q209. Answer: A - works with unlabeled data to find structure or groupings

Rationale: Unsupervised learning uses unlabeled data, seeking to discover structure (e.g., clusters) without predefined target outputs.

Q210. Answer: A - analyzes and interprets text- and voice-based data using the structure of human language

Rationale: NLP uses insight into the structure of human language to analyze and interpret text- and voice-based data.

Q211. Answer: A - machine learning

Rationale: Deep learning is one of the main types of machine learning.

Q212. Answer: A - analysis of large traditional and alternative datasets

Rationale: Fintech-driven analysis of large datasets - both traditional and alternative - is directly relevant to quantitative investment analysis.

Q213. Answer: A - satellite imagery or Internet-of-Things device readings

Rationale: Sensor-generated alternative data includes satellite imagery, IoT readings, and similar machine-collected data.

Q214. Answer: A - social media posts and online reviews

Rationale: Individual-generated alternative data includes social media posts, product reviews, and web search activity.

Q215. Answer: A - help generate alpha and reduce losses

Rationale: Alternative data can be integrated into investment decision-making to help generate alpha and reduce losses.

Q216. Answer: A - data processing and automation of routine tasks

Rationale: Early fintech involved data processing and automation of routine tasks, later advancing to complex machine-learning applications.

Q217. Answer: A - truthfulness, quality, and reliability

Rationale: Veracity concerns the truthfulness/reliability of the data - critical when using Big Data for inference or prediction.

Q218. Answer: A - speed at which data are created and become available

Rationale: Velocity refers to how quickly data are generated and made available; volume is quantity and variety is the diversity of formats.

Q219. Answer: A - range of data types and sources, including structured and unstructured data

Rationale: Variety refers to the diverse types and sources of data, spanning structured and unstructured (traditional and alternative) data.

Q220. Answer: A - hybrid forms of investment decision making

Rationale: These tools affect not only quant managers but also fundamental managers, who use them in hybrid forms of investment decision making.